

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination – 2023

Course: Final Year B. Tech. Branch : AI & Allied Semester : III

Subject Code & Name: BTAIC302 An Introduction to Artificial Intelligence

Max Marks: 60

Date:04-01-24

Duration: 3 Hrs.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) What is AI? Explain the goal of AI.	(Synthesis)	6
B) Explain the Nature of environments with its properties in AI	(Remember)	6
C) What is Intelligent Agent? List the type of Agent and explain any one.	(Understand)	6
Q.2 Solve Any Two of the following.		12
A) Explain steps to solve problem Using AI.	(Synthesis)	6
B) What is a Depth -First Search.	(Understand)	6
C) List search Algorithm and Explain any one of them.	(Remember)	6
Q. 3 Solve Any Two of the following.		12
A) What are Neural Networks? What are the types of neural networks.	(Understand)	6
B) What is the difference between Mutation & Crossover.	(Remember)	6
C) What is ant colony optimization and how does it compare to other methods.	(Synthesis)	6
Q.4 Solve Any Two of the following.		12
A) Explain Recursive Best First Search.	(Synthesis)	6
B) Explain Dijkstra Algorithm.	(Understand)	6
C) What is Conquer Beam Stack Search.	(Remember)	6
Q. 5 Solve Any Two of the following.		12
A) Explain Constraint Propagation.	(Understand)	6
B) What is consistency. Explain Higher order consistency.	(Remember)	6
C) What is Look-head strategies.	(Synthesis)	6

***** End *****